

Pilots Allocation Optimisation for Flinders Port Holdings Final Report

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Project Area: **Pilot Job Demand and Availability Forecasting**
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Abstract

This project final report presents the study on forecasting the pilot job demand and pilot availability in regional ports managed by Flinders Port Holdings, specifically focusing on Whyalla, Port Bonython, Port Lincoln, Port Pirie, and Thevenard. The main objective is to forecast when an additional pilot is required based on historical data and to simulate pilot job demands to predict future daily pilot availability so that the threshold of the current pilot capacity can be found. Utilizing linear regression models, the project forecasts daily pilot job demand and predicts pilot availability over the next five years. The results offer insights into port efficiency, which contribute to the overall performance of the maritime industry.

1 Introduction

1.1 Maritime Operations and Port Dynamics

The maritime industry is an essential part of the global economy, according to the review of marine time transport in 2022, it has revealed that between 2021 and 2022, global marine trade experienced a 3.2% growth, reaching a volume of 11 billion tons.[6] Ports are frequently thought of as global economic hubs, fostering commerce, acting as vital connections in supply chains, giving access to international transportation networks, and fostering regional development. [2, 3, 4, 5]

An integral component of these operations is pilotage. Pilots, after around a year of specialized training, have the responsibility of navigating ships safely within harbours. Their expertise regarding local conditions, including tides, currents, and possible obstructions, is vital for preventing accidents and mitigating environmental hazards.

1.2 Flinders Port Holdings and Regional Ports

Flinders Port Holdings is a leading port operator in South Australia, providing a range of services, including pilotage and mooring. Their commitment to safety and efficiency ensures that vessels are able to navigate and dock without issues at the ports they manage. These ports, namely Whyalla, Port Bonython, Port Lincoln, Port Pirie, and Thevenard, are essential not only for maritime activities but also support various industries, including agriculture, manufacturing, and mining. They facilitate the movement of diverse cargoes such as grains, metals, and petroleum products.

1.3 Objectives

As maritime operations have expanded, so has the complexity of managing pilot schedules and allocations. Traditional methods of allocation, which often relied on heuristics, intuition, and experience, have become less effective in the face of growing demand and dynamic operational constraints. The maritime industry, seeking to minimize delays and manage costs more effectively, needs a more predictive and data-driven approach to tackle this challenge.

This project addresses this gap by diving into the two challenges of forecasting pilot job demand and predicting pilot availability. Using historical data spanning seven years, the research aims to forecast when additional pilots will be required, thereby enabling better scheduling.

By doing so, the project not only seeks to enhance port operation efficiency but also aims to set a benchmark for predictive modelling in maritime operations. While the primary focus is on forecasting and predicting, the research also delves into simulating pilot job demands under varying scenarios. This simulation aids in understanding the scalability and adaptability of the proposed models under different potential future scenarios, from the mundane to the extreme. The importance of this research is underscored by its potential implications: reducing operational costs, enhancing safety, and ensuring timely shipments.

2 Background

Pilot allocation in regional ports has long been a challenge, often leading to inefficiencies and expensive costs. The previous methodologies were either qualitative or lacked real-time data application. With the increased availability of comprehensive data and improved analytical techniques, the current project intends to develop quantitative models for better predicting pilot job demand and availability. The project builds on existing linear regression techniques and specific maritime regulations governing pilot work hours.

2.1 Literature Review

Maritime operations have always been integral to global trade, with regional ports serving as crucial nodes in this vast network. Historically, port operations have relied heavily on a mix of experience-based heuristics and manual interventions to ensure smooth vessel movements.

Dating back to the 1960s, there has been an uptick in the development of models aimed at anticipating shifts in the labour market or proffering strategies for driving desired alterations. In this spectrum, Willems [8] offers insights on various manpower forecasting models. The study predominantly focuses on charting replacement demand, which is best defined as labour demand originating from workers' attrition. Delving into the realm of demand forecasting and its critical role in sculpting informed business strategies, the study by Ubaid, A, Hussain, F and Saqib, M [7] brings attention to an area less travelled, the Australian container shipping sector. Their research contrasted three prominent forecasting models: the Seasonal Auto-Regressive Integrated Moving Average (SARIMA), Holt-Winters' seasonal method, and Facebook's Prophet. The Prophet model distinguished itself not just for its superior accuracy but also for its flexibility in accommodating custom holidays within its forecasting paradigm. Bolat, P and Kayisoglu, G [1] highlighted the importance of effectively managing the high traffic density in the Istanbul Strait. Using historical data from 2012-2019, the ARMA model was developed and subsequently validated through the comparison of real and estimated data. The findings suggest that the ARMA model is suitable for short-term forecasting of vessels navigating the Strait without pilots.

2.2 Current Practices in Pilot Job Allocation

Flinders Port Holdings has identified a need for improved forecasting of pilot job demand in their regional ports to optimize resource allocation,

minimize costs, and maintain operational efficiency. The existing approach is based on real-world situation and choices based on historical data, which could not correctly predict future demand. As a result, a more systematic and data-driven strategy is needed to guide decision-making and guarantee there are enough pilots available when needed. Moreover, according to past pilotage data, it can be found that many pilots showed ‘Rostered On’ but no work is scheduled. At the same time, since pilots need to renew their licenses from time to time and there are training tasks, the specific scheduling of pilots is more complicated, and the cost of adding a pilot is also extremely large.

This project focuses on forecasting the pilot job demand in the regional ports of Whyalla, Port Bonython, Port Lincoln, Port Pirie, and Thevenard. There are currently five pilots servicing these ports, and the goal is to predict when an additional pilot is needed based on past pilotage activities and pilot roster data.

3 Methods

The methods section details the entire strategy used to achieve the two key goals of forecasting pilot job demand and predicting pilot availability. It should be noted that the technique was created to be flexible and reproducible, with an emphasis on data-driven insights and adapting to real-world scenarios.

3.1 Forecasting Pilot Job Demand

3.1.1 Data Collection and Exploration

The study utilized a dataset comprising historical records related to pilot job activities, pilot rosters, and leave data from regional ports, specifically Whyalla, Port Bonython, Port Lincoln, Port Pirie, and Thevenard. Given the timeliness and the incompleteness of the data over the past decade, the data was collected from Flinders Port Holdings and can be traced back to January 2016 till July 2023, so overall with seven financial years of pilot jobs and pilot detailed data for about 6,500 records in history. For pilot job activities, the dataset includes information on port location, vessel movement type (departure, arrival, shift), pilot job start times and end times, job durations, pilot names, and a fatigue level indicator with a corresponding score.

Pilot Job Overview in Regional Ports In order to better understand the current distribution of the number of pilot jobs in regional ports, and subsequently model the prediction of pilot job demands. From the previous analysis, we conducted a thorough analysis of all pilot job data. This comprehensive review led us to group the data based on port, year, and month, providing a clearer picture of the overall demand patterns throughout time. Figure 1 and 2 clearly show a constant growth in the number of pilot jobs over time.

Figure 1: Number of Pilot Jobs vs. Port

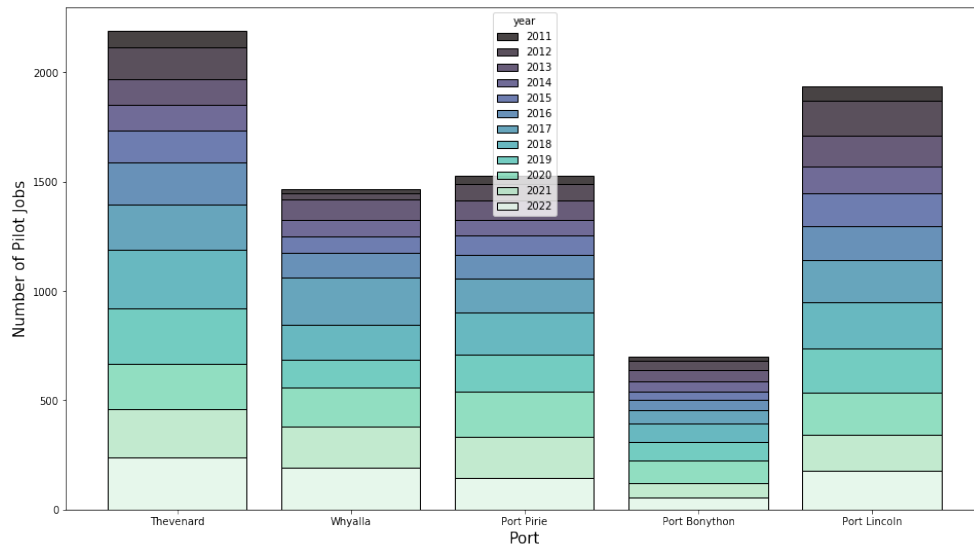
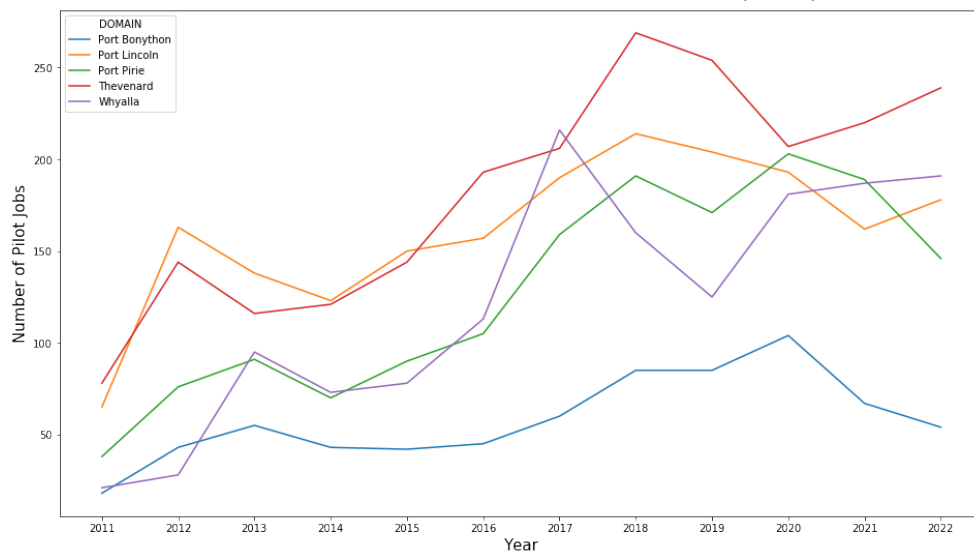


Figure 2: Number of Pilot Jobs vs. Port (Year)

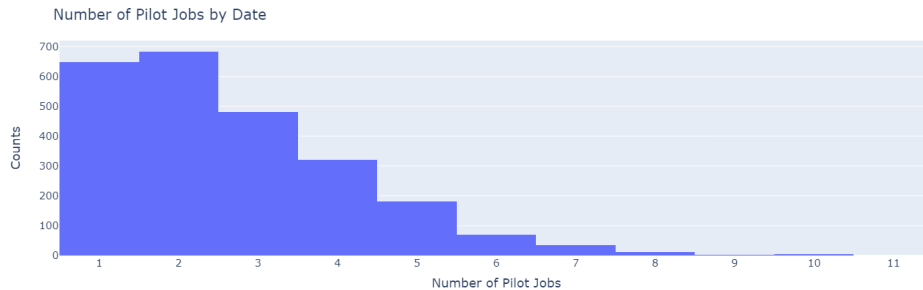


This increase indicates that a growing number of vessels are making their way to the regional ports in South Australia as the years progress. A closer look at annual statistics shows that Thevenard is the port with the most pilot jobs. Port Bonython, on the other hand, ranks last among the five regional ports. A closer examination of Port Bonython's operating details reveals that it predominantly serves tankers transporting gas and petroleum products. The appendix has more information on these statistics. This observed increase in pilot demand for jobs might be linked to increased maritime activities and an improvement in regional

economic prospects. Such a consistent increase in the demand for pilot jobs emphasises the importance of a reliable forecasting approach. We can ensure that the rising demand for pilot services is met smoothly by using accurate predictive measures, so avoiding any negative impacts on port operations.

Since pilot rosters and work schedules are counted in days, we concentrated on counting pilot prior jobs in days to maintain consistency. In more detailed speaking, the past pilot job data has been grouped by day in regional ports, and count the number of jobs per port and per day to find the overall trend. The histogram in Figure 3 shows the statistical distribution of pilot job demand, from which it can be seen that in most cases in the regional ports, about 1 to 3 vessels per day call at the port and need pilotage. Through detailed data analysis, in the five regional ports, there are an average of 2.3 jobs per day, with the number of only 1 pilot job per day accounted for the largest share of 34.57%, the number of 2 pilot jobs per day accounted for 29.49%, and the number of more than 5 pilot jobs per day declined significantly, totalling only 8.12%. In the last 7 years, the maximum number of pilot jobs in a day was 10, and it occurred on 4 days in total, 2018-12-10, 2020-02-23, 2020-03-12 and 2022-03-03.

Figure 3: Statistics on the number of jobs per day



3.1.2 Data Preprocessing

The data was preprocessed before further analysis to assure its quality and usefulness for forecasting models. Several steps were involved in this procedure, including:

- **Data cleaning:** In the past data, we have found that jobs are only recorded if there is a pilot job on that day, so there are instances in the last seven years of data where there is no pilot job on that day, resulting in certain dates being missing, and the entire pilot job count being incomplete. Therefore, the processing of dates was

extended to each day, the number of missing dates was added, and the number of pilot job demands was assigned to 0 for the days when there were no jobs.

- **Data transformation:** The pilot jobs data was converted into a format suitable for time series analysis. Specifically from the pilot job data, we use the variables 'Date', 'Demand' (number of pilot jobs), 'Domain' (port). We have observed that Dates are essentially categorical values and they are in Australia date format (like 'DD-MM-YYYY'). Most machine learning algorithms, including linear regression, expect input features to be numerical. Thus, we need to convert these categorical date values into numerical ones. There are various ways to do this, but a common approach is to convert dates into the number of days since the first date that appears in the past data. With the date in numerical format, the linear regression model can recognize patterns or trends related to the progression of time. If there's a linear trend in 'Demand' over time, the model can capture it using the numerical representation of the date.

3.1.3 Modeling: Linear Regression

Linear regression is a statistical method that models the relationship between a dependent variable and one or more independent variables. In simpler terms, it's like fitting a straight line (or a "trend line") through data points. The goal is to find the line that best describes the observed data.

Linear regression was used to assess the relationship between pilot job demand and date. The model assumes that there is a linear relationship between 'Demand' (the dependent variable - the variable we're attempting to forecast) and 'Date' (the independent variable - the predictor). The data is split into two parts: a training set and a testing set. 80% of the data is used for training the model, and 20% is used for testing its performance. After training, the model then predicts the demand for pilot jobs in the future (based on the test data dates). These predictions are the model's best guess about future demand based on historical trends. The least squares approach was used to estimate the model parameters (coefficients), which minimises the sum of the squared differences between the observed and predicted values.

The linear regression model was evaluated based on its ability to reliably estimate pilot job demand. The predicted values were compared to the actual observed values in the dataset during this evaluation. To assess the accuracy of the models, a range of performance criteria were used, including:

- Accuracy: This statistic measures how well-predicted values match the actual values in the dataset. A greater accuracy implies a better-performing model, whereas a lower accuracy shows that the model may require more development.
- Mean Squared Error (MSE): MSE is a calculation that takes the average of the squared differences between the predicted and true values. The result exposes larger mistakes by squaring the differences, assisting in identifying models that have trouble properly forecasting more extreme values.
- Root Mean Squared Error (RMSE): By reflecting the square root derivation of the MSE, the RMSE gives an error measure articulated in the same unit as the expected values.

3.2 Predicting Pilot Availability

3.2.1 Data Collection and Exploration

It is essential to establish the relationship between pilot job demand and pilot availability in order to decide when an extra pilot is required. Aside from the previously described historical pilot job data, pilot leave data, pilot roster data, and service time configuration data are also utilised to calculate the total hours for completing each work. There are five pilots spread over the five regional ports: one in Whyalla, two in Port Pirie, one in Port Lincoln, and one in Thevenard. The start and end times for each pilot's roster are included in terms of pilot roster data. The pilot leave data captures the type of leave as well as the start and end times of the leave. This extensive data gathering ensures a detailed picture of pilot availability in relation to work demand, enabling in identifying when an additional pilot may be required. Service time configuration gives a detailed time tracking of the pre and post-pilotage job time which includes the travel time between ports, and other documentation time related to pilot jobs.

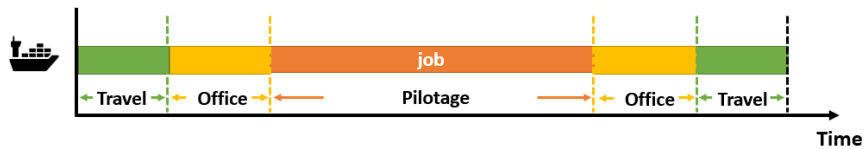
3.2.2 Defining Pilot Availability

The total time for completing a single job for each pilot comprises travel time from home to the job port, pre-pilotage time for the pilot to perform some paperwork at the office, and then the real job duration. After concluding the job, there is time to return to the office, followed by time to return home. So, using the dataset we have, we can determine the entire time period for conducting every job by adding pre/post pilotage

time, actual job duration, and travel time between ports. Figure 4 has give a straightforward illustration of the pilot job timeline.

Figure 4: Illustration of Pilot Job Timeline

• Pilot Job Timeline:



In accordance with Flinders Port Holdings (FPH), pilots operating in regional ports adhere to a scheduling system of "four weeks on, one week off." Importantly, a pilot must not exceed 12 continuous hours of work. Following such an exhaustive shift, they are mandated to have a minimum rest period of 10 hours. For our study, the primary objective is to discern the daily availability of each pilot, governed by the aforementioned regulations. To define pilot availability, let's break down the process step-by-step:

- **Setting the Baseline:** Initially, we assigned the default available time for each pilot as 0 hours.
- **Processing Pilot Job Data:** When a pilot has been assigned a job on a given day, the total work time for that day is subtracted from the 12-hour threshold. The result is the pilot's available time for that specific day.
- **Integrating Pilot Leave Data:** If records indicate a pilot has performed a task while being on leave, his available time is set as the negative of his total work time. This signifies an anomaly or a special scenario. Conversely, if a pilot is on leave and hasn't undertaken any work, his available time remains at 0.
- **Accounting for Pilot Roster Status:** For pilots marked as 'Rostered On' and have work, their available time is calculated similarly to the pilot job data method: 12 hours less the total work time. However, if a pilot is 'Rostered On' but hasn't been assigned any jobs, they naturally have the full 12-hour duration as their available time. Pilots who are 'Rostered Off' and haven't worked will maintain an available time of 0. In the rare occurrence that a pilot is recorded as 'Rostered Off' but has still undertaken work, the available time is represented as the negative of his total work duration for that day.

It's also imperative to ensure that pilots do not breach their operational boundaries. This means that pilots must not exceed the 12-hour work limit and must be given a subsequent 10-hour resting period. An analysis of the historical data reveals that over the past seven years, there were X records where pilots in regional ports exceeded their 12-hour work limit. To rectify this and ensure safety protocols, the subsequent day's availability for such pilots was set to zero, ensuring they receive a mandatory day off.

Building on the previous logic, pilots working at regional ports from 2016 to July 2023 have been extensively measured. Each of these pilots is assigned an indicator called 'Available Time Daily', which represents their vacant hours each day. To get a clearer perspective on the cumulative availability across all regional ports, it's crucial to aggregate this available time for every single day. In doing so, we compile the daily available hours of all pilots, giving insight into the overall capacity that the regional ports possess.

3.2.3 Forecasting Pilot Availability

In order to predict the future availability of pilots to better understand when pilots can be scheduled to work, and thus know the availability of the entire port to determine whether the current number of pilots meets the work demand. We will build a linear regression model to predict daily pilot availability. For this problem, the closely related variables to pilot availability are the number of hours a pilot works per day and the number of jobs, so we will use these two variables as the predictor of the model, and pilot availability as the value to be predicted by the model.

In addition to using pilot job duration as a metric, we also compare the number of jobs a pilot does per day to it, which makes it more intuitive to see that when the number of jobs reaches a certain value, additional pilots are needed to complete the job. When looking at all the regional pilot jobs in the past, we see that there is a significant difference in the pilot job duration, with an average of 1 hour and 47 minutes. The shortest job duration is just 5 minutes, while the longest job lasts 24 hours and 51 minutes. Therefore, the number of pilot jobs is used as an alternative measure to compare pilot available time only for reference, and we mainly focused on finding the relationship between pilot job duration and pilot available time per day.

3.3 Simulation

The simulation process was used to generate synthetic data that closely aligned with the patterns and distribution found in the actual pilot job

demand dataset. By employing this process, it provides an environment to study the potential effects of future pilot demand and availability scenarios without the risks associated with real-world trials.

3.3.1 Define the Objective

The main objective of the simulation was to predict the potential pilot job demand and availability over the next five years, considering two distinct scenarios: the normal case and the extreme case.

3.3.2 Choose Input Distributions

To generate data for the future, the distribution patterns found in the historical dataset were analyzed and modelled. The number of pilot jobs per day, job duration, and other key metrics were considered. The distribution found in the historical dataset was used to model future data.

3.3.3 Define Relationships

The relationships between different variables, as determined by the linear regression model, were established. These relationships provided guidelines for generating synthetic data. For instance, the relationship between date and pilot job demand provided insights into the probable increase or decrease in demand over time.

3.3.4 Generate Data

- Normal Case: Using the relationships and input distributions previously defined, data for the next five years was generated, assuming the continuation of the current trend. This process took into account the yearly increase in pilot jobs, seasonal variations, and other patterns found in the historical dataset.
- Extreme Case: For this scenario, the job demand was artificially increased to be ten times the typical growth rate. This extreme amplification was used to test the system's robustness and adaptability under extreme conditions. While the likelihood of such a scenario occurring is low, it provides insights into potential vulnerabilities and areas that might need reinforcement.

4 Results

4.1 Forecasting Pilot Job Demand

Forecasting of pilot job demand for regional ports is made in general and for individual ports separately.

4.1.1 Trend Analysis

The overall trend derived from the seven-year historical data indicated a modest increase in job demand across the five regional ports. This escalation, while not sharp, was consistent, pointing to the rising significance of efficient pilot allocation.

The relationship between the demand for pilot jobs and the specified variables was shown using the linear regression model. Following pilot availability, the data showed that vessel arrivals or departures had the greatest impact on demand for pilot jobs. This conclusion is consistent with the hypothesis that rising vessel activity would raise demand for pilots.

4.1.2 Model Performance

For all ports in general (Figure 5) For pilot job demand across all regional ports, the linear regression model gives a 42% accuracy considering ‘Date’ as the input. In other words, when the model tries to predict the demand for pilots across regional ports based solely on the date, it’s correct a little less than half the time. Also, the mean squared error is around 3.14, while the root mean squared error is around 1.77.

For individual ports (Figure 6-10) In Table 1, the linear regression model’s outcome for forecasting pilot job demand is displayed. The findings indicate that only Whyalla possesses an accuracy exceeding 60%, with Port Bonython and Port Lincoln marginally above 50%. Nevertheless, the accuracy for the rest of the regional ports appears to be unsatisfactory.

Port	MSE	RMSE	Accuracy
Overall	3.14	1.77	0.42
Whyalla	0.86	0.93	0.64
Port Bonython	0.25	0.50	0.53
Port Lincoln	1.09	1.04	0.53
Port Pirie	0.58	0.76	0.47
Thevenard	0.93	0.96	0.26

Table 1: Linear Regression Prediction Result for Job Demand

Figure 5: Linear Regression Prediction Result in All Ports

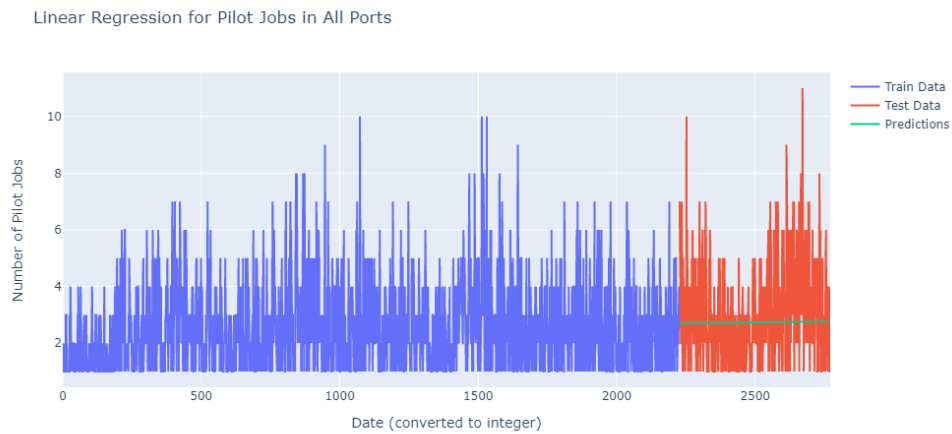


Figure 6: Linear Regression Prediction Result for Whyalla

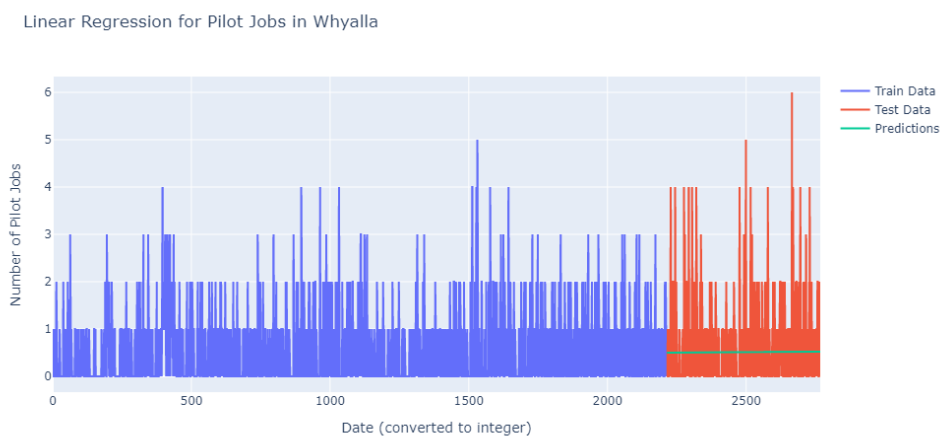


Figure 7: Linear Regression Prediction Result for Port Bonython

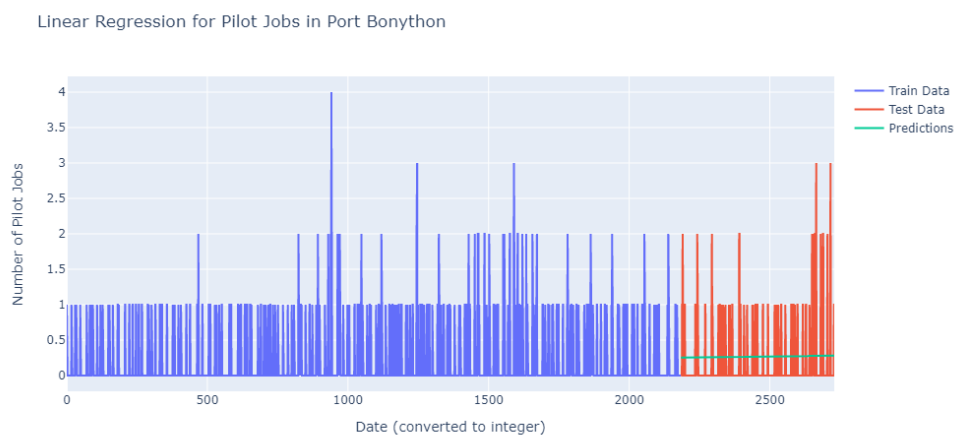


Figure 8: Linear Regression Prediction Result for Port Lincoln

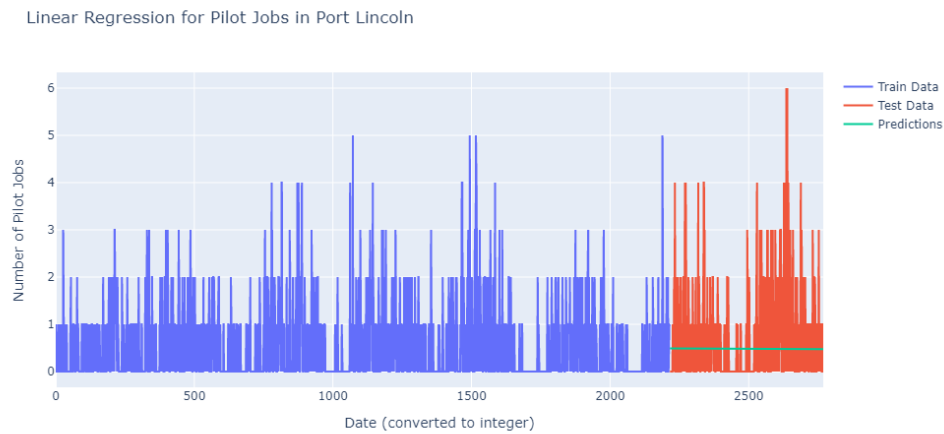


Figure 9: Linear Regression Prediction Result for Port Pirie

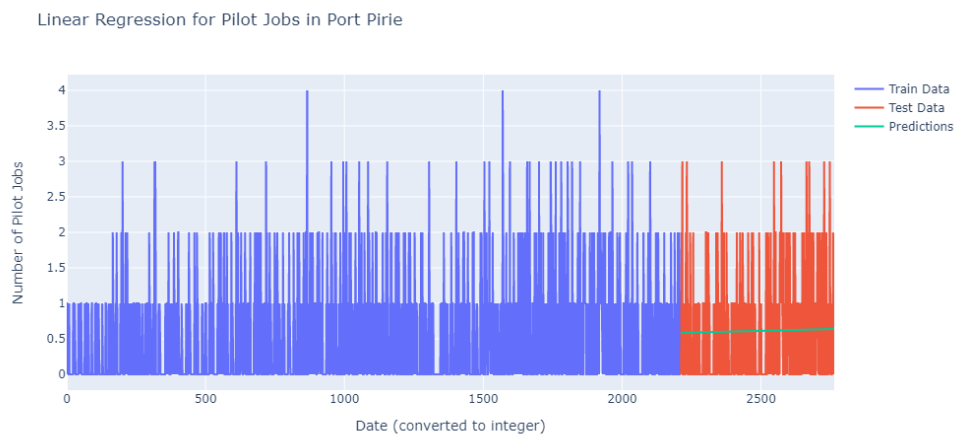
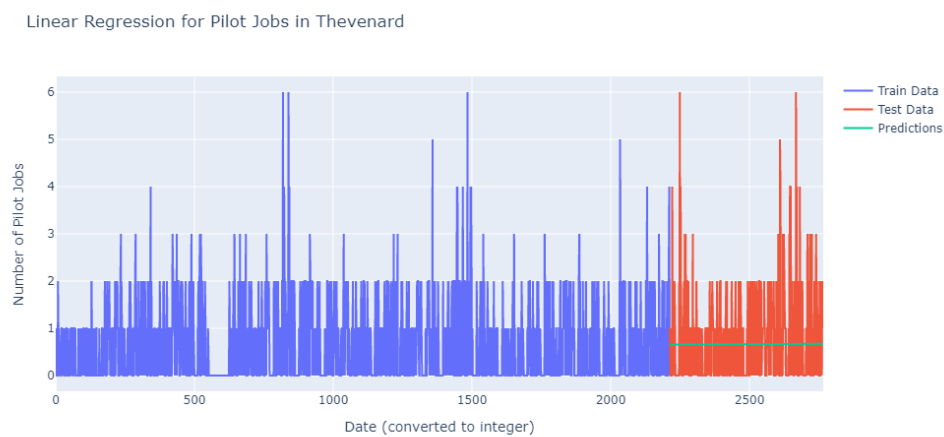


Figure 10: Linear Regression Prediction Result for Thevenard



4.1.3 Predictive Insights

For all ports in general The linear regression model may be stated formally as the following equation,

$$\text{JOB DEMAND} = 0.00013 \times \text{DATE} + 2.42 \quad (1)$$

Specifically, when $\text{DATE} = 0$, which is also the earliest date that exists in our dataset: 2016-01-03 after transformation, the predicted job demand stands at 2.42. For each subsequent day, the job demand increases by 0.00013. If we're looking at this on a daily scale, it means that every day, there's an incremental demand for 0.00013 times the DATE in addition to the base demand. Over a longer period, this small increment can result in a substantial increase. Taking the year as a metric, the increase in demand by the end of the year would be $0.00013 \times 365 = 0.04745$. This translates to an approximate 4.75% rise in demand over the year. This aligns with the observed trends and gives businesses a crucial heads-up to adjust resources, plan schedules, and accommodate the rising demand.

For individual port The prediction trajectory line for individual ports can also be translated into a tangible mathematical equation that reflects the pattern we can observe.

$$\text{Whyalla: JOB DEMAND} = 0.00004 \times \text{DATE} + 0.40 \quad (2)$$

$$\text{Port Bonython: JOB DEMAND} = 0.00005 \times \text{DATE} + 0.15 \quad (3)$$

$$\text{Port Lincoln: JOB DEMAND} = -0.00002 \times \text{DATE} + 0.54 \quad (4)$$

$$\text{Port Pirie: JOB DEMAND} = 0.00011 \times \text{DATE} + 0.35 \quad (5)$$

$$\text{Thevenard: JOB DEMAND} = 0.00003 \times \text{DATE} + 0.59 \quad (6)$$

With the same logic as before, there will be a 1.46% yearly increase based on the initial job demand, pointing towards a modest but steady growth in pilot demand for Whyalla. For Port Bonython, Port Pirie, and Thevenard, the yearly increase will be 1.83%, 4.02%, and 1.10% respectively. Note that only in Port Lincoln, the model suggests a decreasing demand over time. This is crucial as it suggests a potential reduction

in pilot needs for this port, thus, over a year, there will be a decline of 0.73% based on the initial demand.

Based on the above formula and figures, there is a clear reflection that when analysing individual ports, all regression lines converge to a flat straight line and the upward trend appears to be numerically insignificant due to the inability of the available data for the model to capture a clear trend. Therefore, the forecasts of future individual port workloads obtained from the linear regression model for each port are close to the average of the daily job demands up till now.

From the overall model to the prediction model for specific ports, we can clearly see that due to the use of 'Date' as an input variable, which contains a significant amount of outliers and noisy data, our model is unable to find the appropriate pattern from the available data and predict the future specifics with a high degree of accuracy for the prediction of pilot job demand. The outcome can serve as a broad guide, which shows us the average case in the future. This model also validates the observations that historical data suggested a steady increase in pilot job demands. The positive coefficient of DATE indicates an upward trend. Businesses can now be more confident in their planning and resource allocation, knowing that the model's forecasts verify past trends.

4.2 Predicting Pilot Availability

4.2.1 Operational Exceptionals

Exceed Daily Working Limit Delving into the past data, there were 19 instances over the past seven years where pilots in regional ports worked beyond 12 hours per day (0.29% of all regional jobs), with an average working time of around 14 hours.

Rostered Off or On Leave with Jobs Another circumstance exists that pilots sometimes will be assigned jobs when they are 'Rostered Off' or 'On Leave'. It is reasonable for some temporary changes that lead to adjustments in the pilot roster, but sometimes it is not the increased demand for work in the ports that requires more than one pilot, but rather the need for the pilot to always ensure that their licence is valid in each port. So the pilot will also have job assignments during roster off or leave. From the historical data, a total of 434 entries, or 3.13% of the time, pilots have jobs when rostered off or leave during the last seven years, with 20.28% of these occurring during 'Rostered Off'. Pilot on leave with leaving type being 'Travel' has the highest proportion of 62.67%.

As mentioned in the Method section, we set the pilot availability in these cases to a negative value to differentiate between normal working hours. Figures 11 and 12 showed the distribution of different types of negative pilot availability detailing in individual pilots. These exceptions, while few, highlighted the need for thorough examinations and more effective scheduling in the future.

Figure 11: Statistics of Leave Type for Pilot Negative Availability

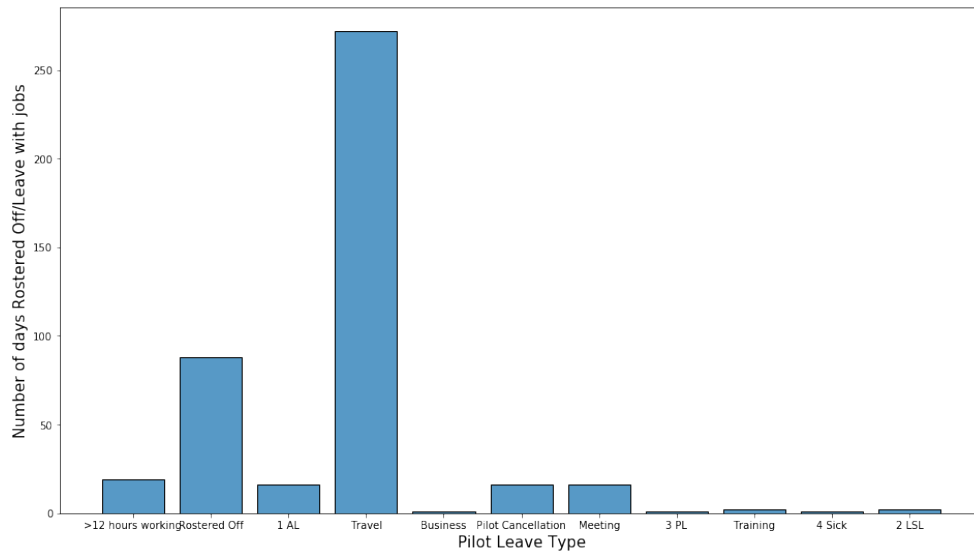
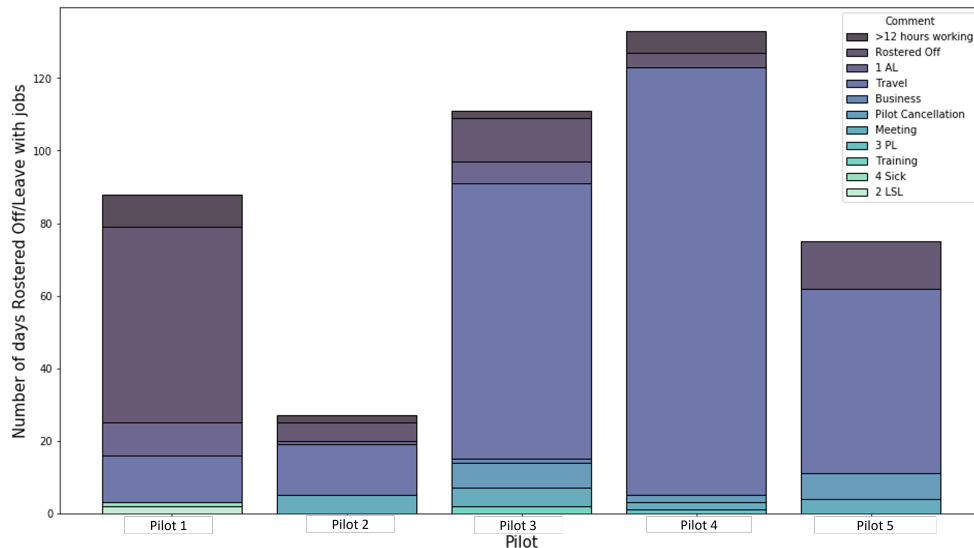


Figure 12: Statistics of Leave Type for Pilot Negative Availability by Pilot



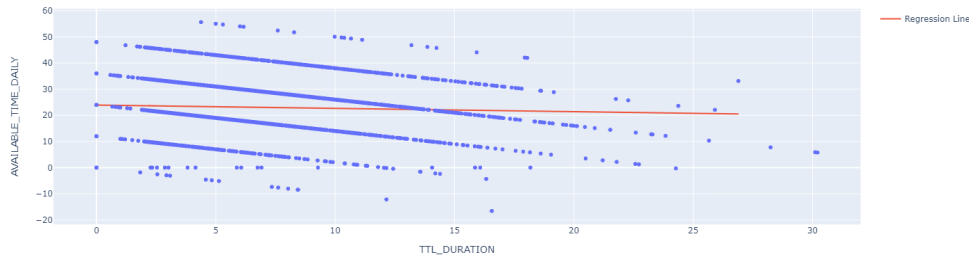
4.2.2 Availability Forecast

Without considering how much time an individual pilot has available, the available time parameter in our model represents the cumulative availability of all the pilots per day. Based on the logic described in the Method Section, we counted pilot availability separately in terms of cumulative job duration and number of jobs, respectively.

Total Job Duration v.s. Pilot Availability Considering there are five regional pilots in total if they are all ‘Rostered On’ on a given day, in the best-case scenario, let’s assume that each of them has an upper limit of 12 hours of availability, then the five of them have a total of 60 hours of availability. Figure 13 has provided a detailed distribution and relationship between job duration and pilot availability, and the regression line can be seen in red, we can formulate the model with mathematics equations:

$$\text{AVAILABLE TIME DAILY} = -0.12 \times \text{TTL DURATION} + 23.89 \quad (7)$$

Figure 13: Linear Regression for Job Duration v.s. Pilot Availability



The model starts with a value of 23.89, which could be interpreted as the total available time in hours that pilots have when there is no accumulated total duration of work ('TTL DURATION' = 0). This suggests that without any assignments or work commitments, pilots have a total available time of approximately 23.89 hours daily across five regional ports. Given that the default availability of each pilot is 12 hours per day, in other words, the model gives us results that indicate that on average 2 regional pilots are rostered on a daily basis.

Additionally, as the amount of work (total job duration) increases, the available time for pilots decreases. This is intuitive in a business context because more work hours would naturally reduce the free time or available time pilots have. For every additional hour of work (1 unit

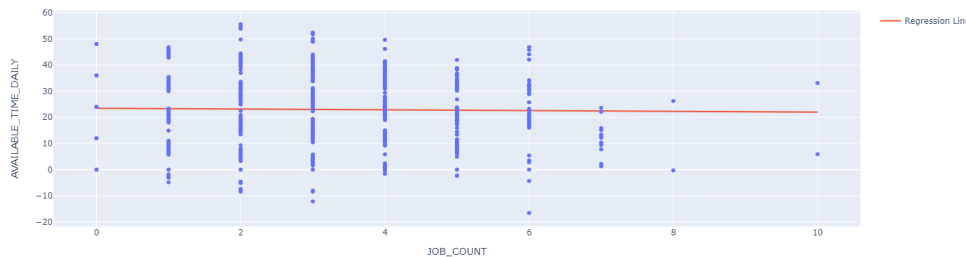
increase in 'TTL DURATION'), the pilots' available time decreases by 0.12 hours, which is approximately 7.2 minutes. If we assume 'TTL DURATION' increases by a constant value every day over the year, then the accumulated decrease in available time daily would be $0.12 \times 365 = 43.8$ hours over a year. Depending on the average duration of work increments, this could mean pilots would be progressively busier as the year advances, leading to a decrease in their total available time by 43.8 hours annually.

Moreover, assuming the current five regional ports have no pilot with free time to take on additional jobs, i.e., 'AVAILABLE TIME DAILY'=0, then the total duration of work is about 192 hours. In other words, when the cumulative length of pilot job demand on a particular day in the five ports is more than 192 hours, the number of the current five pilots and their rosters can not satisfy the job demand, so in this case, an additional pilot is needed to complete the job.

Total Job Counts v.s. Pilot Availability Similar to the previous logic, if we consider the number of jobs as a metric, Figure 14 showed a linear regression model between total job counts and pilot availability. The regression line can be formulated into the equation:

$$\text{AVAILABLE TIME DAILY} = -0.14 \times \text{JOB COUNT} + 23.43 \quad (8)$$

Figure 14: Linear Regression for Job Counts v.s. Pilot Availability



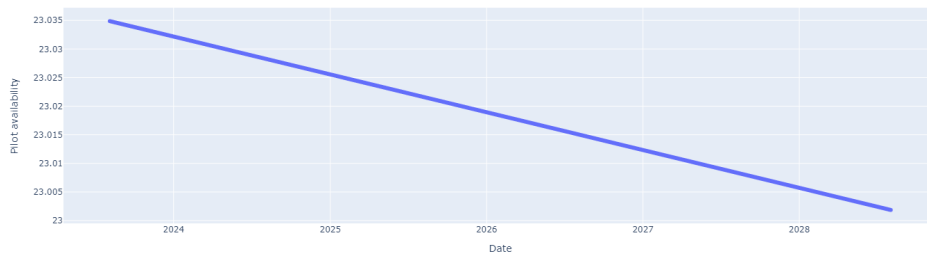
So, without any job, pilots collectively have an available time of roughly 23.43 hours daily across all five regional ports. The coefficient -0.14 for 'JOB COUNT' suggests a decline in daily available time for every additional job assigned. In real scenarios, this indicates that for every new job, the available time for pilots decreases by 0.14 hours, approximately 8.4 minutes. If there's an average increase of 10 jobs daily across all ports, the pilots' total available time would reduce by 1.4 hours daily due to these assignments.

The two models' implications are significant for operations and resource planning. A continual trend of decreasing available time indicates that as the total job duration and job count increase, pilots become busier, which could eventually lead to a shortage of available pilots unless proactive measures are taken. This could mean there exists a potential need to hire more pilots and adjust schedules to ensure job demands are met without overburdening the pilots.

4.2.3 Simulation Outcomes

Normal case scenario The simulated data for the next five years showed pilot availability closely mirroring the predicted job demand. We followed the prediction result of the linear regression model. Since the previous model indicated that job demand tends to increase over time, leading to a decrease in pilot availability. Figure 15 gives a clear illustration of the simulated data and it is clear that the linear downward trend is fully consistent with the model. Thus, the average pilot available time over the next five years for all five regional pilots in five regional ports is around 23.02 hours.

Figure 15: Simulation Normal Case with Linear Model



Extreme case scenario The extreme case, where job demand was multiplied tenfold, revealed a more pronounced downward trend in pilot availability. Figure 16 provides an overview of the simulated data with ten times the current demand. The average pilot available time over the next five years will be decreased to 19.71 hours. These cases, while theoretical, highlighted the potential challenges ports might face in unprecedented surge scenarios and underscored the need for scalable solutions. These two scenarios are only partial simulations of how the port demand might change in the future, while in the meantime, based on the existing model and requirements from FPH, it is possible to continue simulating different scenarios, thus providing a more constructive and possible guide to port operations.

Figure 16: Simulation Extreme Case with Ten Times of Current Distribution



5 Conclusion

The project on Pilot Job Optimization Allocation significantly contributes to the understanding of pilot job demand and availability within regional ports. By utilizing linear regression models and incorporating real-world constraints, the project not only predicts the required metrics but also offers insights into potential future port operations.

For pilot job demand analysis, the seven-year historical data showed a steady, yet modest, increase in the demand for pilot jobs across five regional ports. The linear regression model, using 'Date' as the input, illustrated that vessel activities have the most substantial impact on pilot job demand. For regional ports as a whole, the linear regression model yielded an accuracy of 42%, with individual ports such as Whyalla exceeding 60% accuracy. However, certain ports like Port Lincoln predicted a decline in demand, indicating varied trends across individual ports. The model showed the job demand would rise approximately 4.75% annually across all ports, with individual ports having their respective yearly changes.

For pilot availability analysis, the study highlighted 19 instances over the seven years where pilots exceeded their daily working limit and 434 instances where they were assigned jobs during off times or leaves. Models predicting pilot availability based on total job duration and job count showed a decreasing trend in availability as job demands increased. Simulations indicated that in a scenario with a constant increase in job demand, pilot availability will mirror the demand closely. However, with a tenfold spike in job demand, the average pilot available time over five years drops significantly.

From the previous analysis, a few crucial implications emerge:

- **Increasing Demand:** There's a clear rising demand for pilot jobs across regional ports. This upward trend mandates a proactive approach, from enhancing training programs to potentially recruiting more pilots to handle the rising demands.
- **Efficiency in Scheduling:** The instances where pilots worked beyond limits or during their off times highlight the necessity for better, more efficient scheduling. This not only ensures that pilot needs are met but also guarantees that pilot well-being and safety are prioritized.
- **Variable Trends Across Ports:** As evident from the individual port analysis, not all ports follow the general trend. Thus, a tailored approach, based on specific port demand, is required when strategizing for future needs.

- Getting Ready for Increases: The extreme case simulation underscores the importance of being prepared for potential surges. Ports should have scalable solutions in place to address unprecedented increases in demand.
- Model Limitations: The use of 'Date' as a singular input, especially given its noisiness and outliers, is a limitation. The model, while valuable, serves as a broad guide and might benefit from the inclusion of more relevant variables or a more sophisticated modelling approach.

The conclusions provide a road map for FPH, focusing on proactive planning, increasing operational effectiveness, and taking into account scalable solutions. Ports can assure smooth operations, prioritise pilot safety and well-being, and be well-prepared to handle the future with a strategic vision. Additionally, future studies can explore more complex modelling techniques and incorporate additional variables, such as pilot fatigue level, tide conditions or economic factors, for a more comprehensive analysis.

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In summary, every individual mentioned has significantly contributed to my journey and the completion of this project. I am truly thankful for their support and guidance.

A Appendix

The Appendix section contains a more detailed analysis of pilot job demand, based on different time dimensions (months) and individual ports from the project progress report. Additionally, detailed pilot availability by individual pilots and ports are also shown.

Figure 17: Number of Pilot Jobs in Whyalla grouping by month

Number of Pilot Jobs vs. Year-Month in Whyalla

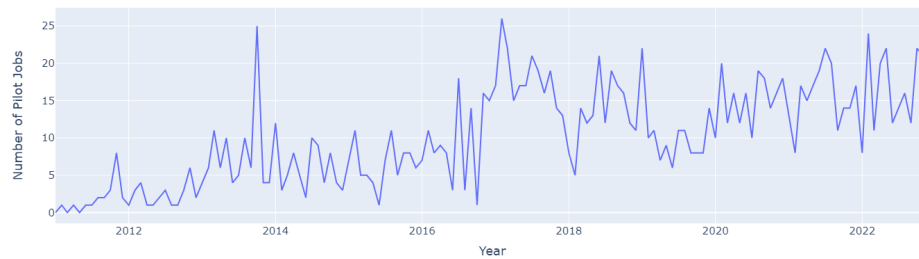


Figure 18: Number of Pilot Jobs in Port Pirie grouping by month

Number of Pilot Jobs vs. Year-Month in Port Pirie

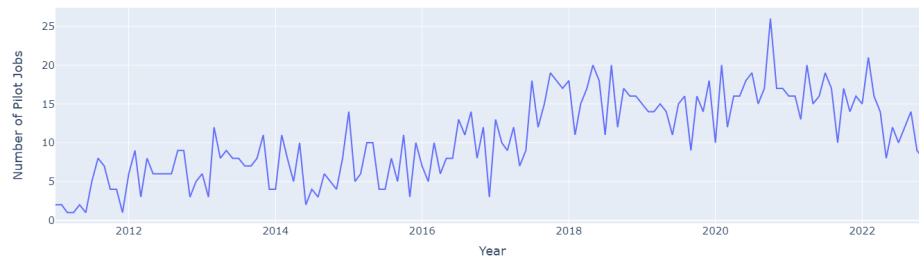


Figure 19: Number of Pilot Jobs in Thevenard grouping by month

Number of Pilot Jobs vs. Year-Month in Thevenard

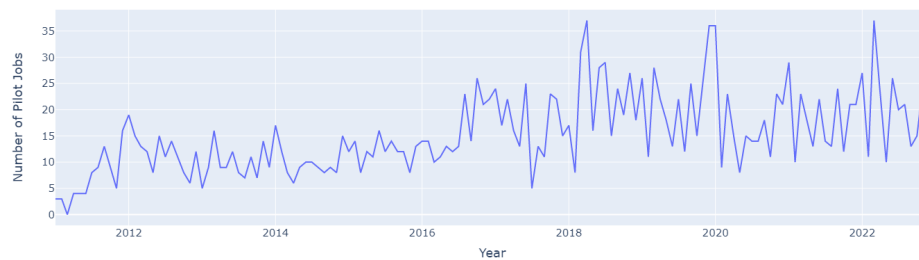


Figure 20: Number of Pilot Jobs in Port Bonython grouping by month

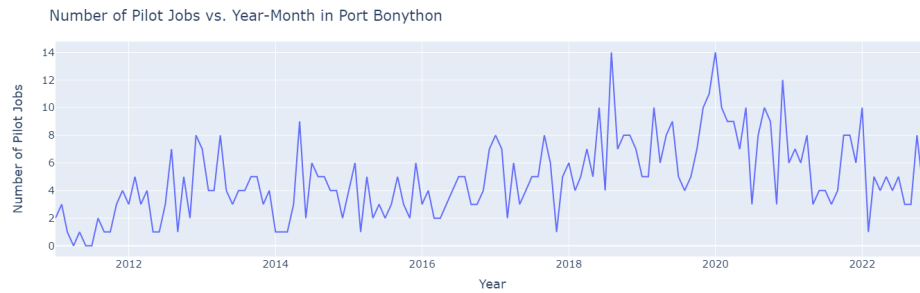


Figure 21: Number of Pilot Jobs in Port Lincoln grouping by month

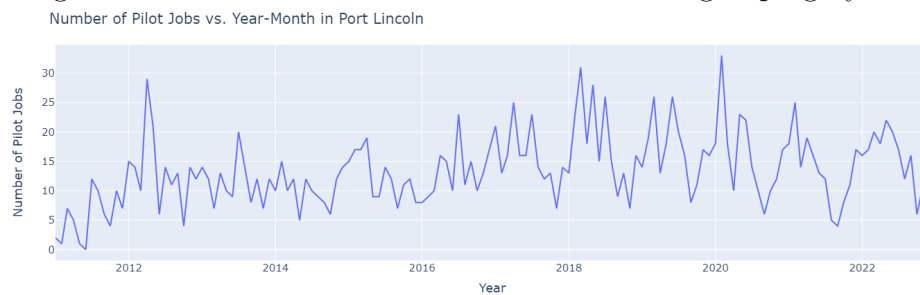


Figure 22: Example of Pilot Roster

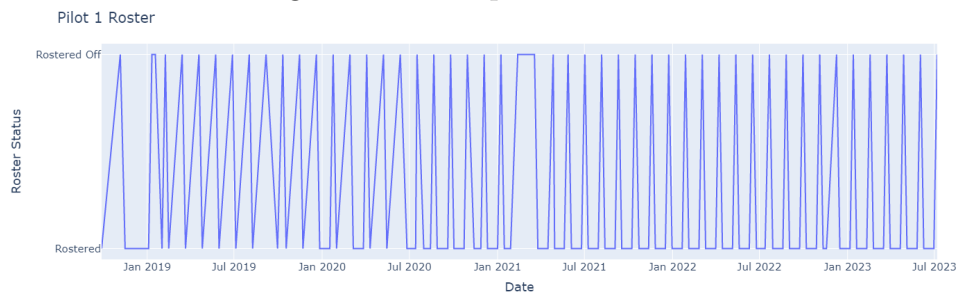


Figure 23: Pilot Available Hours Calculation Logic

Available Time for Each Pilot (hours not working)

Suppose Each job has a maximum of 12 hours working + 10 hours resting

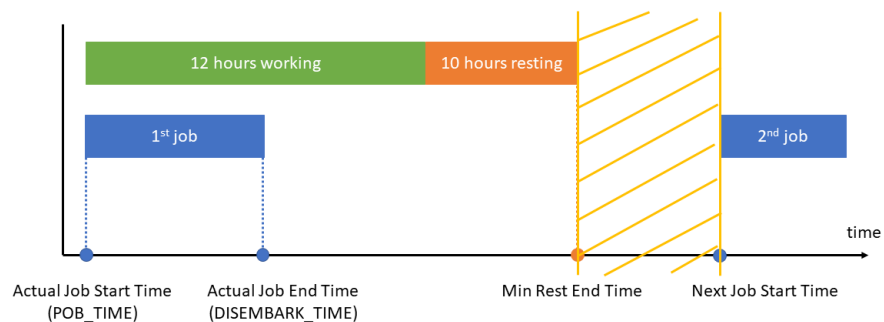


Figure 24: Pilot Availability v.s. Job Duration in All Ports

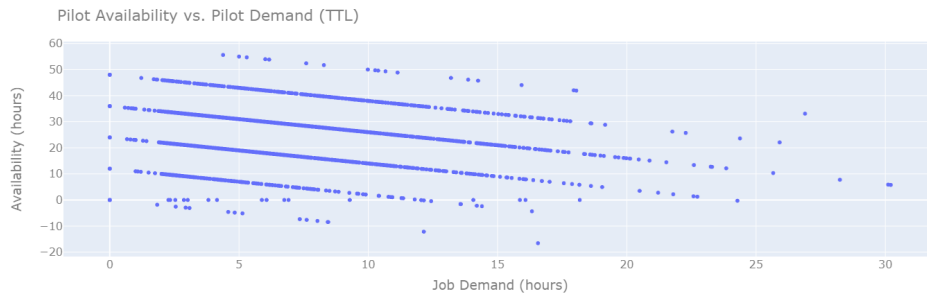


Figure 25: Pilot Availability v.s. Job Duration by Pilots

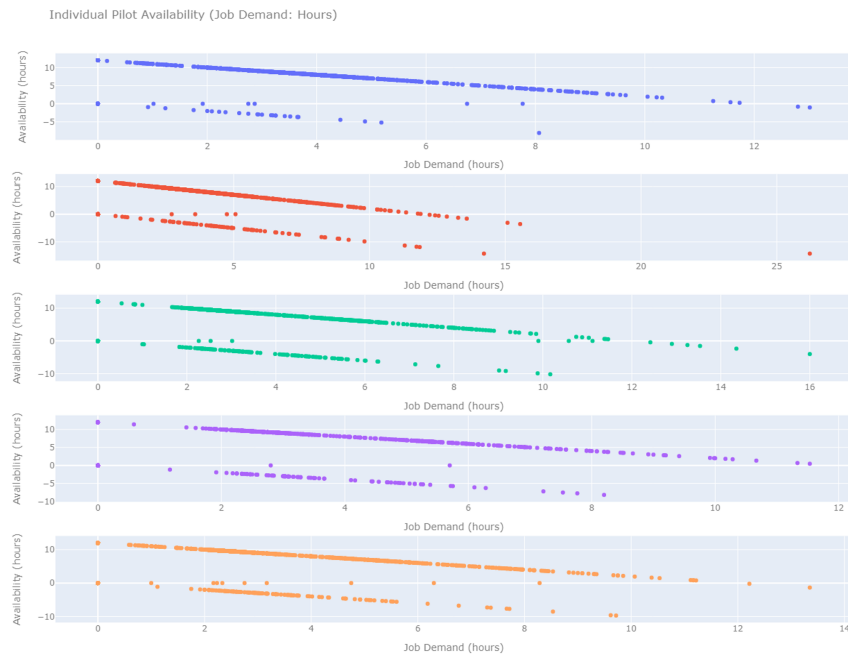


Figure 26: Pilot Availability v.s. Job Duration by Ports

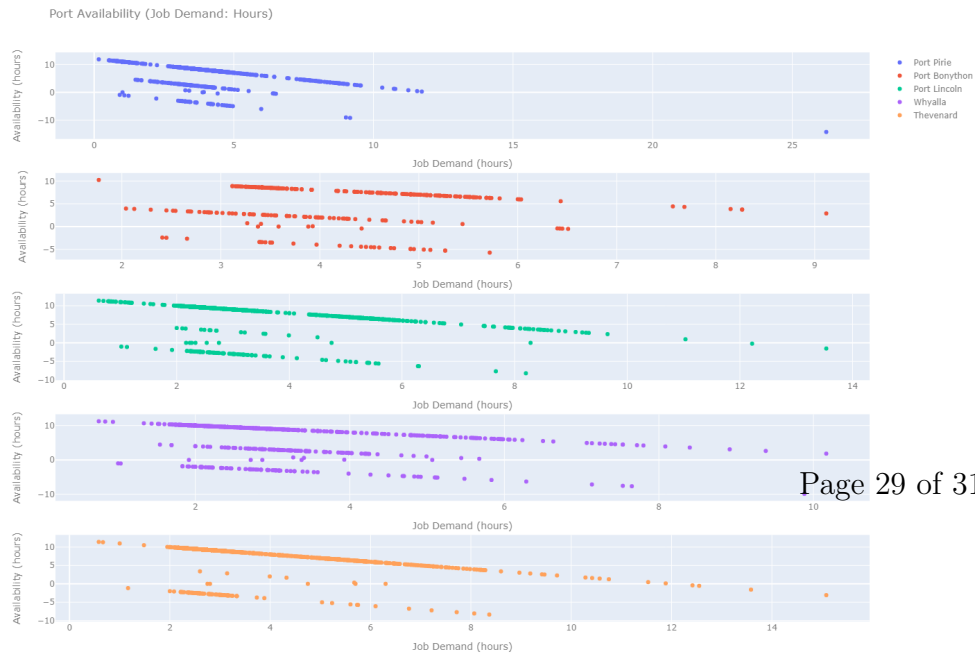


Figure 27: Pilot Availability v.s. Job Counts in All Ports

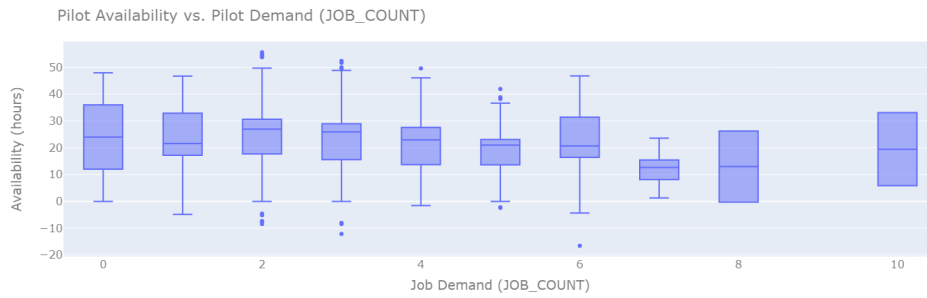


Figure 28: Pilot Availability v.s. Job Counts by Pilots

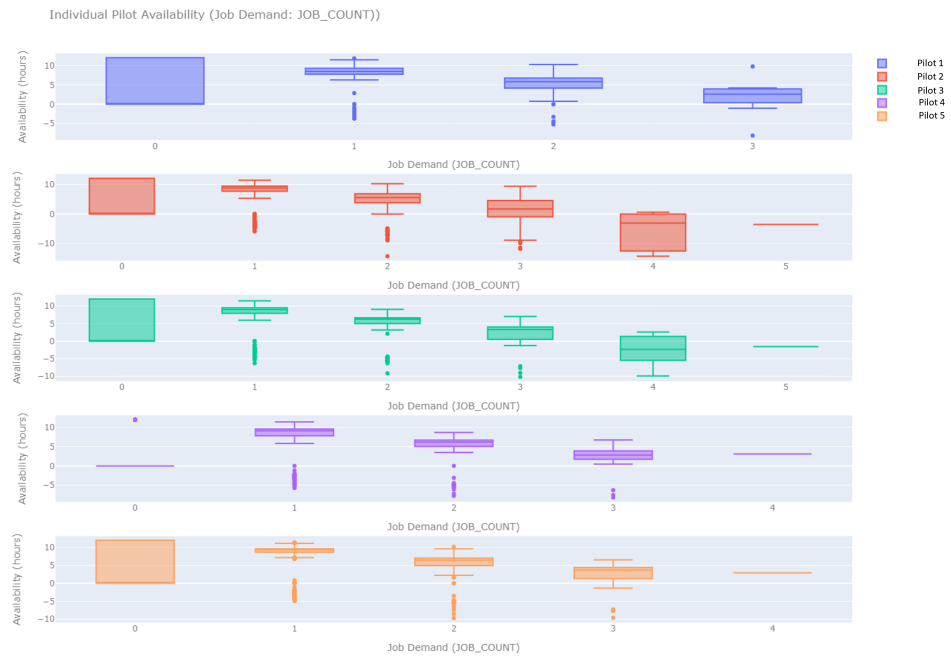
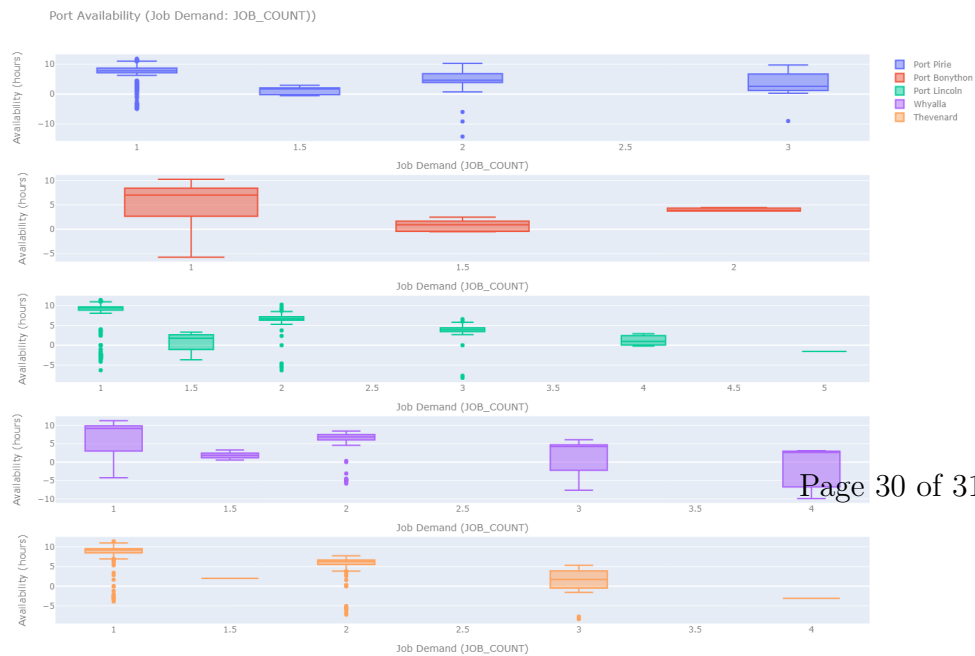


Figure 29: Pilot Availability v.s. Job Counts by Ports



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